BOYA AJAY VARMA

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GitHub: github.com/ajayvarmaboya | LeetCode: leetcode.com/u/ajayvarmab | Portfolio: ajayvarmaboya.github.io/portfolio

Backend & AI Engineer | Open-Source Contributor | Python, Java, FastAPI & LangGraph

PROFESSIONAL SUMMARY

Computer Science (AI & ML) graduate and open-source contributor specializing in scalable backend engineering and document-grounded GenAI architectures. Experienced in developing modular, containerized REST APIs with FastAPI and Spring Boot, orchestrating stateful LangGraph RAG workflows, and building developer automation tools. Strong Git/GitHub fluency (PR workflows, code reviews, issue triage), reproducible Docker environments, and disciplined algorithmic problem solving in Java and Python.

TECHNICAL SKILLS

- **Programming Languages:** Java (17/21), Python, JavaScript, SQL
- **Backend & APIs:** FastAPI, Spring Boot 3, RESTful APIs, JDBC, Pydantic, Uvicorn, OAuth 2.0, Maven
- **AI & Vector Systems:** LangGraph (StateGraph), LangChain, Pinecone, FAISS, Sentence Transformers, RAG Pipelines
- **Open Source & DevOps:** Git & GitHub (PRs, Code Reviews, Actions), Docker, Docker Compose, Linux, Pytest, JUnit
- **Databases & Architecture:** PostgreSQL, MySQL, Schema Design, Relational Modeling, Layered DAO Pattern, Indexing
- **Core Engineering:** Data Structures & Algorithms (DSA), OOP, Error Handling, API Contract Design, Clean Code

SELECTED PROJECTS & OPEN-SOURCE CONTRIBUTIONS

Legixo RAG — Document-Grounded GenAI Pipeline (Primary Code Sample)

Technologies: Python, FastAPI, LangGraph, Pinecone, Groq LLM, Sentence Transformers, Pydantic, Docker

- Architected an asynchronous document QA pipeline using LangGraph StateGraph, orchestrating conditional routing between retrieval relevance validation, source attribution, and response synthesis.
- Integrated Pinecone vector database with Sentence Transformers for semantic chunk retrieval; verified 12/12 automated evaluation test scenarios for hallucination reduction on out-of-domain queries.
- Containerized service using Docker and exposed high-throughput REST endpoints via FastAPI with strict Pydantic schemas, comprehensive error handling, and modular test coverage.

LeetCode GitHub Sync — Open-Source Developer Automation Tool

Technologies: Java, Spring Boot 3, JavaScript, Chrome Manifest V3, GitHub REST API, OAuth 2.0, Maven

- Engineered an automated browser extension and backend microservice that detects accepted submissions and commits structured solution files directly to user GitHub repositories via GitHub REST API.
- Implemented secure GitHub OAuth 2.0 token exchange through a Spring Boot service, integrating Monaco editor DOM extraction, UTF-8 Base64 encoding, and sync retry mechanisms.
- Built validation logic to eliminate duplicate commits, handle wrong-answer edge cases, and log granular sync diagnostics, automating daily developer workflows.

OpenWork — Open-Source Agentic AI Desktop Application

Technologies: Open Source, Python, Docker, Git/GitHub, Linux, Agentic Workflows

- Active contributor to an open-source desktop application powering guided agentic workflows, collaborating with maintainers on issue triage, PR reviews, and code contributions.
- Configured and validated multi-container Docker environments to safely sandbox local autonomous agents and verify end-to-end integration test runs.

JobPrep OS & StockFlow — Layered Backend & Database System

Technologies: Java 17, Spring Boot 3, PostgreSQL, Docker Compose, Maven, Relational Modeling

- Designed relational database schemas with indexed PostgreSQL tables to support interview preparation tracking and order/inventory management workflows.
- Implemented presentation, service, and DAO layers with constructor dependency injection, PreparedStatement SQL injection defense, and custom exception handling.
- Containerized application and database services via Docker Compose for reproducible local setup across development and testing environments.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML)

Sri Venkateswara Engineering College, Tirupati, India | Graduation: 2025 | CGPA: **8.2 / 10**

CERTIFICATIONS & KEY HONORS

- **NPTEL National Certification:** Fundamentals of Cloud, Edge, IoT & ML — **Top 5% National Rank**
- **Algorithmic Practice:** Active problem-solving in Java & Python with structured solutions in public *neetcode-submissions* repository.